

End-to-End Digital Twin System for Cardiac Arrest Risk Detection: Design, Implementation, and Feasibility Evaluation

Zahra Safari
University of Calgary
Calgary, Canada
zar.safari@ucalgary.ca

Emad A.Mohammed
Wilfrid Laurier University
Waterloo, Canada
emohammed@wlu.ca

Behrouz H. Far
University of Calgary
Calgary, Canada
far@ucalgary.ca

Abstract—Sudden cardiac arrest (SCA) remains a leading cause of mortality, accounting for millions of deaths annually worldwide. Yet, most deep learning models for ECG-based risk prediction operate through offline pipelines and are never integrated into continuously operating systems. While digital twin (DT) architectures offer a principled solution by maintaining synchronized virtual representations of patient state, fully operational end-to-end DT implementations integrating live telemetry, cloud-based AI inference, and dynamic twin state management remain largely absent from the literature. This paper presents the design, implementation, and feasibility evaluation of a complete end-to-end digital twin pipeline for near-real-time ECG-based cardiac arrest risk detection. The system integrates simulated ECG telemetry streaming, cloud-based storage, a containerized deep learning inference service, and Azure Digital Twins for continuous state updates, with a web-based dashboard for visualization and automated alerting. The primary contribution is a deployment-ready, cloud-native pipeline architecture that operationalizes automated data ingestion, asynchronous inference, and dynamic twin state updates within a unified modular framework. Under simulated telemetry conditions, the system achieves an end-to-end latency of approximately 20 seconds. The pre-trained deep learning model achieved strong offline performance (accuracy 88%, AUC 0.8851, F1-score 0.8334), and a six-patient pilot confirms stable pipeline execution and consistent alert generation. These results demonstrate the technical feasibility of integrating predictive DL models into an operational DT framework and provide a deployment-ready reference architecture for future real-time cardiac monitoring applications.

Index Terms—Digital Twin, Cardiac Arrest, Deep Learning, Healthcare AI, Risk Detection

I. INTRODUCTION

Sudden cardiac arrest (SCA) is a life-threatening event characterized by the abrupt cessation of heart function, often occurring without warning. Survival rates decline by 7–10% for every minute without treatment [1], and deep learning (DL) models have shown growing promise in predicting cardiac risk from ECG signals [2], [3]. However, most existing approaches operate on static datasets via offline pipelines [4], thereby limiting their applicability in scenarios that require continuous, real-time assessment. Digital twins (DTs) are dynamic virtual representations of physical entities that continuously synchronize with real-world data streams. Originally developed in engineering, DTs are increasingly explored in healthcare for

physiological monitoring and decision support. However, most healthcare DT systems remain at the conceptual or prototype stage [9]–[11], and in cardiology, existing DT approaches focus primarily on simulation or offline analysis [12], [14]. SCA prediction methods remain largely standalone models disconnected from any operational pipeline [17], and no prior work has demonstrated a fully implemented end-to-end DT system integrating live telemetry, cloud-based AI inference, and dynamic twin state updates for cardiac arrest risk detection.

A. Motivation and Contributions

Despite rapid progress in both ECG-based deep learning and digital twin architectures, their integration into a single operational end-to-end pipeline remains an open challenge for cardiac arrest risk detection. Existing cardiac DT systems are either computationally intensive physics-based simulations unsuited for real-time deployment [6], or conceptual frameworks that omit live telemetry and predictive inference [7]. This work is motivated by the need for a concrete, deployment-ready reference implementation that bridges this gap. The primary contribution of this paper is the design, implementation, and feasibility evaluation of a complete end-to-end cloud-native digital twin pipeline for ECG-based cardiac arrest risk detection, from telemetry ingestion through AI inference to dynamic twin state updates and visualization. Secondary contributions include a quantitative evaluation of end-to-end pipeline behavior under simulated telemetry conditions and a modular architecture extensible to other physiological signals and acute conditions.

The remainder of this paper is organized as follows: Section II reviews prior work on digital twin architectures and ECG-based predictive modelling. Section III details the dataset, system architecture, and experimental setup. Section IV presents the results of the end-to-end pipeline evaluation. Section V concludes the paper with a discussion of contributions, limitations, and future research directions.

II. LITERATURE REVIEW

A. Digital Twin Architectures in Healthcare

Digital twins (DTs) have evolved from their engineering roots into promising tools for healthcare applications, enabling virtual representations of patients that dynamically mirror real physiological states. These representations can support continuous monitoring, personalized decision support, and predictive analytics. Foundational work by Minerva et al. [8] outlined the core architectural components of DT systems, including data ingestion, model synchronization, and visualization, establishing a widely adopted technical baseline. Building on this foundation, several systematic reviews have examined the emerging DT landscape in health [9]–[11], identifying common architectural patterns and emphasizing the importance of interoperability, scalability, and validation frameworks. Despite rapid conceptual growth, most healthcare DT systems remain proof-of-concept prototypes that rely on offline synchronization rather than fully operational, continuously updating pipelines.

B. Real-Time Digital Twins

Low-latency pipeline execution is a core design requirement for time-sensitive applications such as cardiac arrest detection, yet this capability remains rare in current DT deployments. Early cardiac DT efforts relied primarily on physics-based electrophysiology models, which provide high mechanistic fidelity but are computationally intensive and poorly suited for real-time inference [12]. Recent developments have introduced hybrid strategies combining data-driven machine learning with physics-informed modelling to accelerate simulation and improve patient-specific adaptation [13]. Efforts to develop population-level cardiac digital twins using machine learning [14] further demonstrate how DTs can support large-scale modelling. Yet most of these frameworks remain focused on offline modelling and lack integration with live telemetry and closed-loop predictive inference, which are essential for operational deployment.

C. Predictive Models and ECG Personalization

In parallel with advances in DL architecture, ECG analysis has undergone a significant transformation for cardiovascular risk prediction. Ribeiro et al. [3] demonstrated the feasibility of end-to-end neural networks for arrhythmia classification on large ECG datasets, establishing a strong foundation for downstream applications. Subsequent research has explored personalized ECG modelling, emphasizing strategies for improving adaptability and generalization across diverse patient populations [15]. More recently, ECG foundation models have emerged, offering robust feature representations adaptable to various cardiac conditions [16]. In the specific context of sudden cardiac death, Kolk et al. [17] provide a comprehensive analysis of AI-based prediction methods, underscoring the need for prospective validation and workflow integration. ECG-centric DT frameworks have also begun to appear. Hu et al. [18] proposed a digital twin generation approach for personalized heart disease detection, though currently limited

to offline evaluation, representing an important step toward integrating predictive models with real-time DT infrastructures.

D. Summary

The reviewed literature reveals rapid advancements in DT technologies and ECG-based deep learning, yet a clear implementation gap persists. Most healthcare DT systems remain at the conceptual or prototype stage [9]–[11], and existing cardiac DT approaches are either computationally intensive physics-based simulations [12] or offline modelling frameworks disconnected from live telemetry [13], [14]. ECG-based predictive models have achieved strong performance in cardiovascular risk prediction [3], [15]–[17], but are typically evaluated in isolation rather than within continuously updating operational systems [18]. Collectively, these observations highlight a gap between what has been proposed in the literature and what has been concretely implemented as a functioning end-to-end pipeline, particularly for cardiac arrest risk detection. This gap is addressed in Section III. Table I summarizes prior studies, outlines their key limitations, and explains how each contributes to or contrasts with our proposed system. This structured comparison highlights the research gaps our work aims to address.

E. Platform Comparison

Most commercial DT platforms, including AWS IoT Twin-Maker [19], Siemens Insights Hub [20], GE Predix [21], and PTC ThingWorx [22], are optimized for industrial assets and lack native support for physiological biosignals and healthcare data standards. Azure Digital Twins was selected for this work due to its graph-based DTDL modelling environment [26], native integration with Azure IoT Hub and serverless functions for scalable telemetry ingestion and dynamic twin state updates, and dedicated healthcare interoperability support, including the Azure API for FHIR and compliance tooling for regulated data handling [23], [24].

III. MATERIALS AND METHODS

A. Dataset

The data and preprocessing pipeline used in this work are adopted from our prior study [30], where two complementary publicly available datasets were combined to improve training diversity and address class imbalance. The MUerte Súbita en Insuficiencia Cardíaca (MUSIC) dataset [27] includes 554 patients with chronic heart failure labeled as either at risk or no risk, with the majority belonging to the no-risk class. The International Cardiac Arrest Research Consortium (I-CARE) dataset [28] contains 413 ICU patients who experienced cardiac arrest and all labeled as at risk. All datasets are publicly available through PhysioNet [29]. In that study, the majority of the data was used for model training and validation, with a held-out test set reserved for evaluation. From this test set, a subset of six patients (three at risk and three no risk) was selected for system-level evaluation in this work. This subset was intentionally chosen to evaluate the behavior of the end-to-end pipeline rather than the statistical performance of the model.

TABLE I
SUMMARY OF RELATED WORKS AND THEIR RELEVANCE

Study	Domain/Focus	Limitations	Relevance to Our Work
Minerva et al. [8]	IoT-based DT architecture	No ML integration; general-purpose	Provides core DT architecture baseline used in our system design.
Katsoulakis et al. [9]	Digital twins in healthcare (review)	Conceptual focus; limited deployment	Highlights gap between conceptual DT systems and operational implementations.
Papachristou et al. [10]	DT advancements and applications in healthcare	No real-time or signal-level analytics	Supports need for operational DT systems beyond conceptual frameworks.
Xames and Topcu [11]	DT systems for healthcare (review)	No cardiac or real-time focus	Supports need for scalable, end-to-end operational pipelines.
Cluitmans et al. [12]	Physics-based cardiac DTs	High computational cost; not real-time	Motivates use of cloud-native ML for low-latency cardiac DT deployment.
Kuang et al. [13]	Hybrid physics-informed DTs	No operational deployment	Inspires combination of ML and cloud infrastructure for real-time DT updates.
Qian et al. [14]	Population-scale cardiac DTs	Offline simulation; no live telemetry	Demonstrates ML-driven cardiac DT feasibility; motivates operational extension.
Ribeiro et al. [3]	Deep learning for ECG arrhythmia classification	Offline model; no DT integration	Provides pre-trained model weights reused as the inference backbone in our system.
Ding et al. [15]	Personalized ECG modelling	Offline evaluation; no telemetry	Motivates personalized inference as a future extension of our pipeline.
Tian et al. [16]	ECG foundation models	Pre-training only; not deployed in DTs	Informs model selection for robust and transferable ECG feature extraction.
Kolk et al. [17]	AI for sudden cardiac death prediction	No operational DT or pipeline	Establishes SCA prediction as an unmet need within operational DT systems.
Hu et al. [18]	ECG-based DT for heart disease detection	Offline DT generation; no live telemetry	Closest ECG-DT prior work; highlights absence of real-time inference and twin state updates.

Specifically, it enables verification of key system properties, including latency, synchronization between components, and correct alert generation. The balanced composition ensures that both at-risk and no-risk scenarios are represented, allowing validation of both alert triggering and normal-state handling. Additionally, this scale allows detailed inspection of system behavior, including digital twin updates, inference timing, and dashboard synchronization. To simulate real-time operation, historical ECG segments from these patients were streamed repeatedly at fixed intervals. The original preprocessing pipeline from [30] was preserved, including bandpass filtering (0.5–50 Hz), resampling to 400 Hz, and segmentation into 10-second windows. At this stage, a larger cohort was not needed since the goal was to verify that the system works correctly, not to measure statistical model performance.

B. Deep Learning Inference and Integration

The DL model component of the system is a ResNet-based deep neural network for binary classification of SCA risk, adopted from our prior work [30]. The model was originally pre-trained on over 2 million 12-lead ECG recordings from the Brazilian Telehealth Network [3] for six-class arrhythmia classification, and subsequently fine-tuned via transfer learning to accept six-lead ECG inputs and perform binary SCA risk classification; full training details, architecture specifications, and evaluation results are provided in [30]. The trained model was exported in .h5 format and integrated into a containerized inference service running continuously inside

a Docker container hosted on an Azure virtual machine. Instead of processing live patient data, the inference service operates on simulated telemetry generated from historical ECG segments. Every five seconds, the service polls Azure Digital Twins by monitoring changes to each twin’s `lastUpdated` property. When a new update is detected, the corresponding ECG segment is retrieved from Azure Blob Storage via the `ecgBlobUri` field and passed to the model for inference. Prediction results are written back to the digital twin by updating the `riskStatus`, `warningFlag`, and `lastUpdated` properties, enabling continuous and asynchronous twin state updates without manual intervention. This design enables seamless integration of a pre-trained model into a monitoring pipeline without requiring additional retraining.

C. Digital Twin Architecture

The proposed system leverages a digital twin architecture to support continuous monitoring of patient states and risk classification. Each physical patient is represented by a corresponding digital twin, hosted on Microsoft Azure and modelled using the Digital Twins Definition Language (DTDL). These virtual representations maintain structured metadata about the patient’s state and synchronize with incoming telemetry updates. While the current prototype utilizes a single device to simulate telemetry for multiple patients, the system design supports horizontal scaling, enabling each patient to be connected to their own wearable device in future implementations. Each digital twin contains a standardized set of properties that are

continuously updated as new data arrives. These include the `patientId`, the latest `ecgBlobUri` linking to the ECG signal segment stored in Azure Blob Storage, the `lastUpdated` timestamp, the computed `riskScore`, the `riskStatus` indicating whether the patient is currently “Normal,” “At Risk,” or “Pending,” and a binary `warningFlag` used to trigger alerts. The raw ECG waveform data is stored externally, and only secure URIs are maintained in the twin to reduce memory overhead and support flexible querying. Telemetry ingestion is performed by uploading a new ECG segment for each patient every 20 seconds. These uploads are routed through Azure IoT Hub, which sends a message to an Azure Function. The function locates the correct digital twin based on the patient ID and updates its metadata fields, including `ecgBlobUri`, `lastUpdated`, and signal parameters. This metadata change triggers the prediction service to run inference. The prediction service is deployed as a containerized microservice built using Docker. A pre-trained ECG classification model and supporting libraries are packaged into the container and hosted on Azure infrastructure. Every five seconds, the service polls all digital twins and checks the `lastUpdated` field. If a change is detected, it retrieves the corresponding ECG segment from Blob Storage, applies preprocessing, and performs inference. The model outputs a continuous risk score ranging from 0 to 1. If the score exceeds the predefined threshold of 0.5, the twin’s `riskStatus` is updated to “At Risk”, and the `warningFlag` is set to true. Otherwise, the status remains “No Risk”. This system has a built-in notification and alert system. Whenever a PatientTwin transitions to an at-risk state, the corresponding update triggers an automated email alert that includes the patient’s identifier, timestamp, and risk-score information. This ensures that clinicians or system operators are notified immediately, even if they are not actively viewing the dashboard. At the same time, the dashboard highlights the affected patient and displays a warning message to provide clear visual feedback. Figure 1 shows the Azure Digital Twins Explorer interface used to visualize the state of each patient twin. In this example, six patient twins (P0004 through P0057) are active and displayed in a spatial graph layout. The right-hand panel shows the properties of the selected twin, including the URL of the latest ECG segment, the timestamp of the last update, and the current status fields. At system initialization, all twins are set to “pending” with a risk score of zero, indicating that telemetry ingestion has begun but inference has not yet occurred. Once a new ECG segment is processed, these values are updated to reflect the classification result.

The proposed system is intended for continuous monitoring of patients who are not yet under continuous hospital monitoring. In such scenarios, wearable ECG devices can stream data to the system, where the digital twin maintains an up-to-date representation of patient state. When an elevated risk is detected, the system generates alerts that can prompt clinician review or further diagnostic assessment. This positions the framework as a decision-support tool for early risk indication rather than immediate emergency intervention. By leveraging the standardized DTDL schema, the twin model can be ex-

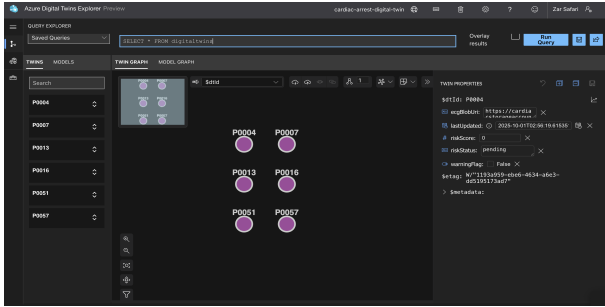


Fig. 1. Azure Digital Twins Explorer

tended in the future to incorporate additional physiological indicators or to support more advanced notification channels (such as SMS, mobile push alerts, or cross-system integrations) without requiring major architectural modifications. The current system sends an automated alert when risk is detected; defining the appropriate response workflow is left for future clinical validation work. Overall, the digital-twin layer provides a flexible, scalable, and event-driven foundation that supports both real-time monitoring and timely delivery of risk alerts.

D. Dashboard Visualization

To visualize the results, a web-based dashboard was developed and deployed on Azure App Service, providing a clinician-facing interface for monitoring patient states and risk classifications. The dashboard automatically refreshes every 10 seconds, aggregating information from the digital twin layer. Key components include a semicircular gauge chart summarizing the overall system risk, a pie chart displaying patient distribution by risk category, a bar chart showing individual patient risk scores, and a scatter plot mapping heart rate against predicted risk. These visualizations allow clinicians to assess both population-level trends and individual patient status quickly. A patient table below these summaries lists all monitored patients, their heart rates, risk classifications, and last update times, along with a “View” button for accessing patient-specific details. Figure 2 illustrates the main dashboard interface. Each visualization panel reflects the most recent state stored in the digital twins, ensuring consistency between back-end inference and front-end monitoring.

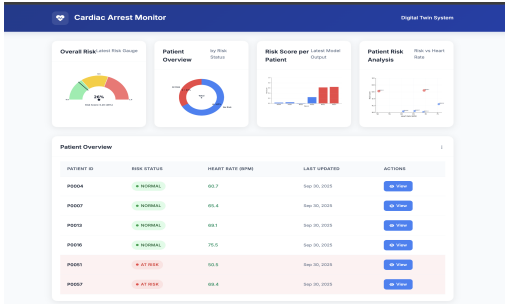


Fig. 2. Dashboard overview displaying system-wide and patient-level risk distribution and trends.

The patient detail page (Figure 3) displays metadata, including patient ID, risk status, heart rate, and the last updated timestamp. A semicircular gauge visualizes the most recent risk score, and ECG waveform plots for three leads are displayed with QRS peaks highlighted in red. These plots enable rapid assessment of signal quality and model output interpretation.

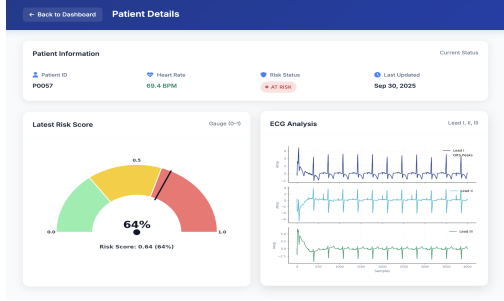


Fig. 3. Patient detail page displaying ECG waveforms with QRS annotations and corresponding risk summaries.

Figure 4 provides an overview of the full system pipeline, including telemetry ingestion, digital twin updates, machine learning inference, dashboard refresh, and alert propagation. This pipeline illustrates the system’s modular structure, where ingestion, inference, and visualization operate asynchronously but remain synchronized through the digital twin layer.

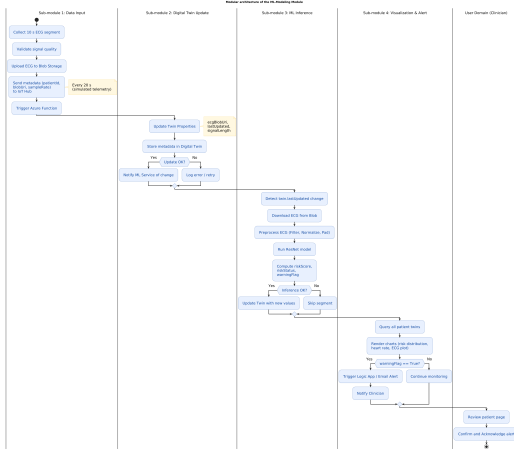


Fig. 4. End-to-end architecture of the digital twin system. ECG telemetry is simulated every 20 seconds, inference is triggered every 5 seconds, and the dashboard refreshes every 10 seconds.

The digital twin layer provides several advantages over a direct telemetry-to-dashboard pipeline. It ensures persistent, queryable state management for each patient, decouples ingestion, inference, and visualization components to enable independent scaling, and uses the standardized DTDL model to support future integration of additional data streams (e.g., blood pressure) or external applications (e.g., mobile alerts) without major structural changes.

IV. RESULTS AND DISCUSSION

The proposed end-to-end DT system was evaluated using simulated telemetry streams from six patient twins (three no-risk and three at-risk), drawn from the MUSIC and I-CARE test sets described in Section III-A. Each twin received new ECG segments every 20 seconds, which were processed by the deployed deep learning model for SCA risk classification. Prediction results were stored in the digital twin and visualized through the monitoring dashboard. The system achieved an overall accuracy of 83.3% across the six-patient pilot. All no-risk patients were correctly classified, while two of the three at-risk patients were correctly identified. One at-risk case was misclassified as no-risk, resulting in perfect specificity with reduced sensitivity. Compared to the offline model accuracy of 88% (AUC: 0.8851, F1-score: 0.8334), the end-to-end system shows a modest decrease of 4.7%. This difference is expected in a feasibility prototype and is likely due to operational factors such as asynchronous telemetry ingestion, polling intervals, and timing misalignment between system components. In addition, the small sample size limits statistical generalizability, as the goal of this evaluation is system-level validation. Overall, the pipeline demonstrates consistent predictive behavior under deployment conditions. This evaluation focuses on system-level feasibility rather than clinical validation. The system demonstrated stable end-to-end operation throughout the experiment. Telemetry ingestion, digital twin updates, inference triggering, and dashboard synchronization all operated reliably without failure. The measured end-to-end latency, from ECG upload to digital twin update, ranged from 13 to 25 seconds, with an average latency of approximately 20 seconds. This includes telemetry ingestion and twin metadata update (5 s), inference polling and ECG retrieval (5 s), model inference and digital twin property update (5 s), and dashboard refresh (10 s, occurring in parallel with inference and update operations). Automated alerts were successfully triggered for all at-risk cases, demonstrating that risk predictions were correctly reflected in the digital twin state and consistently propagated to both the alerting mechanism and the visualization dashboard.

In the context of ambulatory cardiac monitoring, a 20-second end-to-end latency is appropriate for continuous risk surveillance. Unlike defibrillation scenarios where immediate response is required, the proposed system is designed for pre-arrest risk stratification by identifying patients trending toward elevated risk over longer monitoring periods. For this use case, sub-minute alert generation may provide earlier awareness compared to periodic manual review or episodic clinical assessment.

Overall, these results validate the technical feasibility of integrating a pre-trained deep learning model within a cloud-based digital twin pipeline. They also highlight how real-world system constraints influence model behavior, emphasizing the importance of evaluating AI models within their deployment context. In real-world settings where cardiac events are rare, the rate of false alerts will need to be evaluated and reduced in future work to avoid unnecessary interruptions.

V. CONCLUSION, LIMITATION, AND FUTURE WORK

This paper presented the design, implementation, and feasibility evaluation of a complete end-to-end cloud-native digital twin pipeline for ECG-based cardiac arrest risk detection. The system integrates simulated ECG telemetry streaming, a containerized deep learning inference service, Azure Digital Twins state management, and a monitoring dashboard within a unified modular architecture. The primary contribution is a deployment-ready reference implementation that bridges the gap between conceptual digital twin architectures and fully operational systems, validated through quantitative latency measurement and a controlled six-patient pilot study. Results confirm stable pipeline execution, consistent alert generation, and predictive performance closely aligned with offline evaluation, demonstrating that end-to-end deployment conditions have a measurable but contained effect on model behavior.

The current prototype has several limitations that define the scope of this feasibility study. The evaluation relied on simulated telemetry from a small cohort of six patients using replayed ECG data rather than live device streams, limiting generalizability. The model follows a generalized rather than personalized training approach, and the system relies solely on ECG signals without multimodal physiological inputs. The fixed classification threshold and absence of temporal pattern tracking limit the sophistication of the alerting mechanism. The current alert mechanism does not include a defined clinical escalation protocol. These constraints are intentional boundaries of a system-level feasibility study, with clinical validation and large-scale testing explicitly deferred to future work.

Future work will focus on four directions. First, integration with live wearable ECG devices will enable genuine real-time telemetry evaluation and robustness testing in uncontrolled environments. Second, latency reduction through event-driven architectures, parallelized inference, and adaptive thresholding will improve pipeline responsiveness and alert precision. Third, incorporating multimodal physiological inputs, patient-specific fine-tuning, and larger-scale evaluation across diverse populations will enhance prediction robustness and clinical generalizability. Fourth, defining and validating the appropriate response workflow following an alert will be essential for real-world clinical deployment.

REFERENCES

- [1] D. A. Berg, R. M. Sutton, et al., "Association between time to defibrillation and survival in pediatric in-hospital cardiac arrest with a first documented shockable rhythm," *JAMA*, vol. 326, no. 18, pp. 1795–1804, 2021.
- [2] J. M. Kwon, et al., "Deep learning-based algorithm for detecting cardiac arrest using electrocardiogram data," *npj Digit. Med.*, vol. 3, no. 1, p. 64, 2020.
- [3] A. H. Ribeiro et al., "Automatic diagnosis of the 12-lead ECG using a deep neural network," *Nature Communications*, vol. 11, no. 1, p. 1760, 2020.
- [4] H. Rajput, et al., "Transfer learning-enabled heart disease detection using digital twins," *IEEE Access*, vol. 11, pp. 15846–15857, 2023.
- [5] R. Minerva, G. Lee, and N. Crespi, "Digital twin in the IoT context: A survey on technical features, scenarios, and architectural models," in *Proc. IEEE Comput. Commun. Workshop Conf. (CCWC)**, 2020, pp. 1–6.
- [6] J. Li, et al., "A physiological digital twin for cardiac electrophysiology simulation," *Front. Physiol.*, vol. 15, 2024.
- [7] N. Yousef, et al., "Digital twin for intelligent context-aware IoT healthcare systems," *Sensors*, vol. 23, no. 7, p. 3537, 2023.
- [8] R. Minerva, G. M. Lee, and N. Crespi, "Digital Twin in the IoT Context: A Survey on Technical Features, Scenarios, and Architectural Models," *Proceedings of the IEEE*, vol. 108, no. 10, pp. 1785–1824, 2020.
- [9] E. Katsoulakis et al., "Digital twins for health: a scoping review," *npj Digital Medicine*, vol. 7, 2024.
- [10] K. Papachristou, G. Moustakas, and D. Tzovaras, "Digital twins' advancements and applications in healthcare: Towards precision medicine," *Personalized Medicine*, 2024.
- [11] M. D. Xames and T. G. Topcu, "A systematic literature review of digital twin research for healthcare systems," *IEEE Access*, 2024.
- [12] M. J. M. Cluitmans et al., "Digital twins for cardiac electrophysiology: state of the art and future challenges," *Frontiers in Physiology*, 2024.
- [13] K. Kuang et al., "Med-Real2Sim: Non-invasive medical digital twins using physics-informed self-supervised learning," *NeurIPS/ArXiv preprint*, 2024.
- [14] S. Qian et al., "Developing cardiac digital twin populations powered by machine learning provides electrophysiological insights," *Nature Cardiovascular Research*, vol. 4, no. 5, pp. 624–636, 2025.
- [15] C. Ding et al., "Advances in deep learning for personalized ECG diagnostics: a systematic review," *Biosensors & Bioelectronics*, vol. 271, 117073, 2025 (Epub 2024-12-16).
- [16] Y. Tian et al., "Knowledge-enhanced ECG foundation model for universal cardiac disease detection," *Cell Reports Medicine*, vol. 5, no. 8, 2024.
- [17] M. Z. H. Kolk et al., "Prediction of sudden cardiac death using artificial intelligence: Current status and future directions," *Heart Rhythm*, vol. 22, no. 3, pp. 756–766, 2025 (Epub 2024-09-06).
- [18] Y. Hu et al., "Personalized heart disease detection via ECG digital twin generation," *arXiv preprint*, arXiv:2404.11171, 2024.
- [19] Amazon Web Services, "AWS IoT TwinMaker – Create digital twins of real-world systems," 2024. [Online]. Available: <https://docs.aws.amazon.com/iot-twinmaker/>. (Accessed: May 15, 2024).
- [20] Siemens, "Insights Hub: Industrial IoT as a Service," 2024. [Online]. Available: <https://www.siemens.com/en-us/products/insights-hub/>. (Accessed: May 15, 2024).
- [21] GE Digital, "Predix Platform: Asset Performance Management," 2024. [Online]. Available: <https://www.governova.com/software/documentation/predix-platforms/redirect-platforms.html>. (Accessed: May 15, 2024).
- [22] PTC, "ThingWorx Platform: Industrial IoT and Augmented Reality," 2024. [Online]. Available: <https://www.ptc.com/en/support/help/thingworxdocsresources>. (Accessed: May 15, 2024).
- [23] Microsoft, "Azure Policy regulatory compliance controls for HIPAA and HITRUST," 2024. [Online]. Available: <https://learn.microsoft.com/azure/governance/policy/samples/hipaa-hitrust>. (Accessed: May 15, 2024).
- [24] Microsoft, "Azure API for FHIR: Overview," 2024. [Online]. Available: <https://learn.microsoft.com/azure/healthcare-apis/azure-api-for-fhir/overview>. (Accessed: May 15, 2024).
- [25] Amazon Web Services, "AWS HealthLake: Purpose-Built for Health Data," 2024. [Online]. Available: <https://aws.amazon.com/healthlake/>. (Accessed: May 15, 2024).
- [26] Microsoft, "Azure Digital Twins: Overview," 2024. [Online]. Available: <https://learn.microsoft.com/azure/digital-twins/overview>. (Accessed: May 15, 2024).
- [27] Martin-Yebra, A., Martínez, J. P., Laguna, P. (2025). MUSIC (Sudden Cardiac Death in Chronic Heart Failure) (version 1.0.1). PhysioNet. RRID: SCR007345. <https://doi.org/10.13026/z3m7-rf58>
- [28] Amorim, E., Zheng, W., Lee, J. W., Herman, S., Ghassemi, M., Sivaraju, A., Gaspard, N., Hofmeijer, J., van Putten, M. J. A. M., Reyna, M., Clifford, G., Westover, B. (2023). I-CARE: International Cardiac Arrest REsearch consortium Database (version 2.1). PhysioNet. RRID: SCR007345. <https://doi.org/10.13026/m33r-bj81>
- [29] PhysioNet, "PhysioNet: A resource for complex physiologic signals," MIT Laboratory for Computational Physiology, 2024. [Online]. Available: <https://physionet.org/>. (Accessed: May 15, 2024).
- [30] Safari, Z., Mohammed, E., Far, B. (2025). Transfer Learning for Multi-Lead ECG Analysis in Cardiac Arrest Risk Classification. Proceedings of the Canadian Conference on Artificial Intelligence. Retrieved from <https://caiac.pubpub.org/pub/kx8er0ej>